AWS OPERATIONS INTELLIGENCE AI

Enterprise Operations Control Center (EOCC)

Executive Proposal

Executive Summary

Modern enterprises have invested heavily in operational tools such as AWS, Splunk, Jira, ServiceNow, Confluence, GitHub, and Microsoft Teams.

While these systems provide valuable information, operational knowledge remains fragmented across multiple platforms.

During incidents, engineers often spend significant time gathering information before they can begin solving the actual problem.

Common questions include:

- Who owns this service?
- What business process is impacted?
- Which teams are affected?
- Has this issue occurred before?
- Is there a runbook available?
- What is the approved recovery process?
- Which manager should be notified?

The information exists, but it is scattered across different systems.

AWS Operations Intelligence AI is designed to solve this problem by creating an Enterprise Operations Control Center that continuously connects operational data, ownership information, documentation, incidents, architecture knowledge, and business context into a single intelligence platform.

The goal is not to replace existing tools.

The goal is to create an intelligence and control layer above existing enterprise investments.

What Problem Are We Solving?

Today:

```
AWS
Splunk
Jira
Confluence
Teams
GitHub

↓

Disconnected Information
```

Result:

- Slow investigations
- Ownership confusion
- Knowledge silos
- Repeated incidents
- Increased operational risk

Proposed Solution

Enterprise Operations Control Center

The platform continuously:

- Monitors enterprise systems
- Understands relationships
- Investigates incidents
- Coordinates teams
- Provides recommendations
- Supports controlled operational actions

Core Product Capabilities

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1. Enterprise Ownership Intelligence

Problem:

During incidents, teams often do not know who owns a service.

Feature:

Instant ownership discovery.

Example:

Question:

Who owns Glue Job CN3?

Response:

- Owning Team: CNS Team
- Tech Lead: John Smith
- Manager: Jane Doe
- Escalation Channel: Teams Channel Link

Business Value:

Reduces ownership confusion and escalation delays.

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2. Business Process Intelligence

Problem:

Operational tools show infrastructure failures but not business impact.

Feature:

Business process mapping.

Example:

Question:

What business capability is impacted?

Response:

- Customer Onboarding
- New Account Creation
- Customer Verification Workflow

Business Value:

Helps leadership understand business impact immediately.

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3. Enterprise Dependency Intelligence

Problem:

Teams do not know what will break when a service fails.

Feature:

Dependency analysis.

Example:

Question:

What happens if Lambda A fails?

Response:

- Customer API impacted
- Reporting Service impacted
- Analytics Pipeline impacted

Business Value:

Faster impact assessment and decision making.

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4. AI Incident Investigation

Problem:

Engineers manually collect information from multiple systems.

Feature:

Automated investigation.

Example:

Incident:

Glue Job Failure

Platform automatically gathers:

- Splunk logs
- Jira history
- Runbooks
- Ownership information
- Dependencies

Business Value:

Reduces investigation effort significantly.

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5. Historical Incident Intelligence

Problem:

Organizations repeatedly solve the same problems.

Feature:

Incident memory.

Example:

Question:

Has this happened before?

Response:

Incident INC-2345

Previous Root Cause: Database Connection Pool Exhaustion

Previous Resolution: Connection Pool Configuration Updated

Business Value:

Leverages organizational knowledge.

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6. Enterprise Knowledge Search

Problem:

Knowledge exists but is difficult to find.

Feature:

Unified enterprise search.

Example:

Question:

How do we recover Order Processing?

Platform searches:

- Confluence
- Runbooks
- Jira
- Teams

Business Value:

Faster access to operational knowledge.

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7. Architecture Intelligence

Problem:

Architecture knowledge is difficult to access.

Feature:

Architecture understanding.

Example:

Question:

Show architecture of Customer Onboarding.

Response:

- API Layer
- Lambda Services
- SQS
- Glue
- RDS

Business Value:

Accelerates troubleshooting and onboarding.

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8. Operational Risk Detection

Problem:

Operational risks remain hidden.

Feature:

Risk identification.

Example:

Platform identifies:

- Missing owners
- Missing runbooks
- Outdated documentation

Business Value:

Improves operational maturity.

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9. Ownership Gap Detection

Problem:

Some applications have no clear owner.

Feature:

Ownership governance.

Example:

Platform identifies:

- 17 applications without ownership records

Business Value:

Improves accountability.

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10. Runbook Coverage Analysis

Problem:

Critical services may lack recovery procedures.

Feature:

Runbook governance.

Example:

Platform identifies:

- Customer Reporting Service has no recovery runbook

Business Value:

Improves resilience.

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11. Organizational Dependency Intelligence

Problem:

Cross-team dependencies are difficult to understand.

Feature:

Team relationship mapping.

Example:

Question:

Which teams depend on the Platform Team?

Response:

- Payments Team
- Customer Team
- Analytics Team

Business Value:

Improves operational planning.

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12. Expert Dependency Detection

Problem:

Critical knowledge may exist only with one engineer.

Feature:

Knowledge concentration analysis.

Example:

Platform identifies:

- 82% of Glue incidents require one engineer

Business Value:

Reduces organizational risk.

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13. Change Risk Intelligence

Problem:

Deployment risks are difficult to predict.

Feature:

Pre-deployment analysis.

Example:

Question:

What could be impacted by this deployment?

Response:

- Customer Onboarding
- Reporting Pipeline

Business Value:

Safer deployments.

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14. Executive Incident Briefings

Problem:

Leadership needs business-focused updates.

Feature:

Automated executive summaries.

Example:

Response:

Customer onboarding degraded.

Estimated impact:

- New customer registrations affected
- Two dependent teams impacted

Business Value:

Improves leadership visibility.

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15. Enterprise Search Across Systems

Feature:

Single search experience.

Example:

Search:

Customer Onboarding

Results include:

- Jira Tickets
- Confluence Pages
- Runbooks
- Architecture Documents

Business Value:

Reduces information retrieval effort.

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16. Governance Intelligence

Feature:

Operational governance dashboard.

Example:

Displays:

- Ownership Coverage
- Documentation Coverage
- Runbook Coverage

Business Value:

Operational maturity visibility.

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17. Operational Digital Twin

Feature:

Live enterprise relationship model.

Example:

Business Process ↓ Application ↓ AWS Service ↓ Team ↓ Manager

Business Value:

Enterprise-wide situational awareness.

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18. Controlled Operational Actions

Feature:

Execute approved actions directly from the platform.

Example:

Restart ECS Service

Workflow:

AI Recommendation ↓ Human Approval ↓ Execution ↓ Audit Logging

Business Value:

Faster recovery with governance.

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19. Autonomous Incident Coordination

Feature:

Coordinate incident response activities.

Example:

Platform automatically:

- Creates Jira Incident
- Identifies Owners
- Notifies Teams
- Shares Runbooks

Business Value:

Reduces coordination effort.

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20. Enterprise Operations Command Center

Feature:

Single operational view.

Displays:

- Business Processes
- Applications
- Ownership
- Risks
- Incidents
- Dependencies

Business Value:

Unified enterprise operations visibility.

Why This Is Different

Traditional tools focus on:

- Logs
- Metrics
- Dashboards
- Tickets

AWS Operations Intelligence AI focuses on:

- Ownership
- Dependencies

- Knowledge
- Governance
- Business Processes
- Operational Coordination
- Controlled Execution

The platform transforms operational information into operational intelligence.

Expected Business Benefits

- Reduced investigation effort
- Faster incident resolution
- Improved ownership visibility
- Reduced knowledge silos
- Improved governance
- Reduced operational risk
- Improved operational maturity
- Better executive visibility
- Faster onboarding of engineers
- Better cross-team collaboration

Proposed Next Step

Phase 1 Pilot

Focus Areas:

- Ownership Intelligence
- Dependency Intelligence
- Incident Context Generation
- Enterprise Search
- Operations Command Center

Success Criteria:

- Reduced investigation effort
- Improved ownership visibility
- Faster access to operational knowledge
- Increased operational transparency

The long-term vision is to establish a centralized Enterprise Operations Control Center that continuously improves operational efficiency, governance, resilience, and decision-making across the organization.